General Regulations.

- Submission is optional but recommended; tutors provide feedback. Exercises are not graded.
- Work in groups of two or three, preferably from the same tutorial group. Submit all solutions as a single combined PDF via the Übungsgruppenverwaltung, once per group, and include all group members' names.
- Solutions to tasks marked "solve yourself, without an AI agent" may be handwritten and scanned or typeset using LaTeX. Please ensure that scans are legible and not blurry.
- Course material is available in the GitHub repository.

1 LIGO glitches: cross-validation and leakage

Solve parts (a), (c) and (e) yourself, without an AI agent. Use an agent to write the code in parts (b) and (d).

The LIGO detectors record short bursts of instrumental and environmental noise, called *glitches*, that can mimic or mask gravitational-wave signals. The Gravity Spy project sorts glitches into classes by their time–frequency images, using human-classified examples.[1] The folder `sheet04/data` contains 6866 labelled samples in 18 classes, recorded by the Hanford (H1) and Livingston (L1) detectors during LIGO's first observing run, between September 2015 and January 2016, in the file `gravityspy_glitches.csv`. For each glitch, a trigger pipeline provides six numbers: duration, peak frequency, central frequency, bandwidth, amplitude and signal-to-noise ratio.

In this exercise we want to predict the class of a glitch from these six numbers using k-nearest neighbours (k-NN). We want to find the optimal value for k using *K-fold cross-validation* (which allows to use all the data for both training and validation):

1. shuffle the samples and split them into K folds of (nearly) equal size, the same folds for all values of k;
2. for each k in a fixed set of trial values and for each fold $m = 1, \dots, K$, fit the model on the other $K-1$ folds and compute its error e_m on fold m, the fraction of misclassified glitches;
3. for each k, compute $\mathrm{CV} = \frac{1}{K}\sum_{m=1}^{K} e_m$.

In the end we choose the value for k that produced the model with the smallest CV error.

(a) Why do we need a validation set, here the held-out fold, when we tune the hyperparameter k of k-NN? Hint: Consider the training error of 1-NN.

(b) Choose $k \in \{1, 3, 5, 11, 21, 51, 101\}$ by 5-fold CV with randomly assigned glitches, using all glitches in the file (ignore the column `sample_type`). As features, use the standardized logarithms

$$z_{ij} = \frac{\log_{10} x_{ij} - \mu_j}{\sigma_j},$$

where x_{ij} is feature j of glitch i, and μ_j and σ_j are the mean and standard deviation of $\log_{10} x_{ij}$ over the glitches in the training folds only. Plot the CV error against k and state the value of k you choose.

(c) The causes of glitches, such as ground motion during a storm, can last from minutes to days, so several glitches in the dataset can originate from the same long-lasting event. You want a classifier for the

[1] M. Zevin et al., Class. Quantum Grav. **34**, 064003 (2017). Data: S. Coughlin, Gravity Spy Training Set, doi:10.5281/zenodo.1486046 (2018), CC BY 4.0; we use the glitches of the first observing run, without the single example of the class 1080Lines and without the classes 1400Ripples, Helix and Violin_Mode, which occur almost only on a single day.

glitches of the coming weeks of the same observing run. Will random 5-fold CV estimate its error on those glitches correctly, too optimistically or too pessimistically? Why?

(d) Repeat the CV of (b) with three ways of assigning glitches to folds: each glitch at random, as in (b); all glitches of the same day in the same fold, with the days assigned to folds at random; and the same with weeks instead of days. Number the days and weeks by $\lfloor t/86400 \rfloor$ and $\lfloor t/604800 \rfloor$, where t is the column `event_time`, in seconds. The folds will then differ in size; still average the fold errors as above. Plot the CV error against k for all three.

(e) Compare the results with your prediction in (c). Which of the three CV errors, at which k, would you report as the error to expect in the coming weeks? What does this number still not capture?

2 Two routes to the Gaussian-process prediction

Solve this exercise yourself, without an AI agent.

Let $f \sim \mathcal{GP}(0, k)$, observed without noise, $y_i = f(\boldsymbol{x}_i)$, $\underline{\boldsymbol{y}} = (y_1, \ldots, y_n)^\mathsf{T}$. The observations and the value $f(\boldsymbol{x})$ at a new input $\boldsymbol{x}$ are jointly Gaussian, with mean zero and

$$\begin{gathered} \operatorname{Cov}(\underline{\boldsymbol{y}}) = \boldsymbol{K}, \qquad \operatorname{Cov}\big(\underline{\boldsymbol{y}}, f(\boldsymbol{x})\big) = \underline{\boldsymbol{k}}_{\boldsymbol{x}}, \qquad \operatorname{Var} f(\boldsymbol{x}) = k(\boldsymbol{x}, \boldsymbol{x}), \\ \underline{\boldsymbol{k}}_{\boldsymbol{x}} = \big(k(\boldsymbol{x}_1, \boldsymbol{x}), \ldots, k(\boldsymbol{x}_n, \boldsymbol{x})\big)^\mathsf{T}. \end{gathered} \tag{1}$$

(a) We predict $f(\boldsymbol{x})$ by a weighted sum of the observations,

$$\hat{f}(\boldsymbol{x}) = \underline{\boldsymbol{\lambda}}_{\boldsymbol{x}}^\mathsf{T} \underline{\boldsymbol{y}} = \sum_i \lambda_i(\boldsymbol{x})\, y_i, \tag{2}$$

and choose the weights that minimize the expected squared error

$$\mathbb{E}_f\big[\big(f(\boldsymbol{x}) - \underline{\boldsymbol{\lambda}}_{\boldsymbol{x}}^\mathsf{T} \underline{\boldsymbol{y}}\big)^2\big], \tag{3}$$

where $\mathbb{E}_f$ is the expectation over the prior, $f \sim \mathcal{GP}(0, k)$. Show that the optimal weights are

$$\underline{\boldsymbol{\lambda}}_{\boldsymbol{x}} = \boldsymbol{K}^{-1} \underline{\boldsymbol{k}}_{\boldsymbol{x}}, \tag{4}$$

which give the estimate

$$\hat{f}(\boldsymbol{x}) = \underline{\boldsymbol{k}}_{\boldsymbol{x}}^\mathsf{T} \boldsymbol{K}^{-1} \underline{\boldsymbol{y}}. \tag{5}$$

(b) The minimum of the expected squared error (3) is called the kriging variance $s^2(\boldsymbol{x})$. Show that

$$s^2(\boldsymbol{x}) = k(\boldsymbol{x}, \boldsymbol{x}) - \underline{\boldsymbol{k}}_{\boldsymbol{x}}^\mathsf{T} \boldsymbol{K}^{-1} \underline{\boldsymbol{k}}_{\boldsymbol{x}}. \tag{6}$$

(c) Parts (a) and (b) found the best linear predictor by minimizing the expected squared error. We now take a Bayesian perspective and show, for the example of a single observation ($n = 1$; the general case is analogous), that it leads to the same result: condition the joint Gaussian of $f_1 = f(\boldsymbol{x}_1)$ and $f_2 = f(\boldsymbol{x})$ on the observation $f_1 = y$. The pair has mean zero and covariance

$$\begin{pmatrix} k_{11} & k_{12} \\ k_{12} & k_{22} \end{pmatrix}, \qquad k_{11} = k(\boldsymbol{x}_1, \boldsymbol{x}_1), \quad k_{12} = k(\boldsymbol{x}_1, \boldsymbol{x}), \quad k_{22} = k(\boldsymbol{x}, \boldsymbol{x}).$$

The posterior $p(f_2 \mid f_1 = y) \propto p(y, f_2)$ is the slice at $f_1 = y$ through the joint Gaussian of (f_1, f_2). Complete the square in f_2 in its exponent to read off the posterior mean and variance. Show that they equal Eqs. (5) and (6) for $n = 1$: the best linear predictor is also the posterior mean. Hint:

$$\begin{pmatrix} a & b \\ b & d \end{pmatrix}^{-1} = \frac{1}{ad - b^2} \begin{pmatrix} d & -b \\ -b & a \end{pmatrix}.$$

3 Interactive exploration of Gaussian processes

Solve this exercise yourself, without an AI agent.

Open the lecture's demo, "Three values of a Gaussian process",

`mlph-08-gp-explorer-demo.html`

(download, open in any browser, offline). Find out what each panel shows, but do not move anything yet.

The demo uses the Matérn covariance of the lecture, with smoothness ν, range ℓ and signal variance $\sigma_f^2 = 1$: for two sites a distance d apart,

$$k(d) = \frac{2^{1-\nu}}{\Gamma(\nu)}\, u^{\nu} K_{\nu}(u), \qquad u = \frac{\sqrt{2\nu}\, d}{\ell},$$

where K_ν is the modified Bessel function of the second kind. For example, $k(d) = e^{-d/\ell}$ for $\nu = 1/2$, $k(d) = (1 + \sqrt{3}\, d/\ell)\, e^{-\sqrt{3}\, d/\ell}$ for $\nu = 3/2$, and $k(d) = e^{-d^2/2\ell^2}$ for $\nu \to \infty$. The hyperparameter σ^2 is the nugget: measurements at the sites t_i are $y_i = f(t_i) + \varepsilon_i$ with independent noise of variance σ^2.

Answer the following questions before touching the controls. Afterwards, play with the demo, check your answers and explain any errors you made.

Assume that two of the three sites are observed.

(a) When you move the *location* of a measurement in the top right plot, will the Gaussian in the top left plot change? Will the weight functions $\lambda_i(t)$ change?

(b) When you keep the measurement location fixed but move the *observed value* up or down, will the shape of the Gaussian in the top left plot change? Will the weight functions change?

(c) How will the expected squared error in the bottom right plot change as you modify the hyperparameter ν or ℓ?

(d) How will the Gaussian-process prediction, the blue line in the top right plot, change as you introduce a nugget effect with the hyperparameter σ^2 in the bottom left plot?